

# PROJECT TRICORDER Mk I

## Complete Guide to Features, Capabilities, and Theory of Operations

Software 0.6.32 · 15 September 2026

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*RENT-A-HAL Android Edition. Companion to the Field Operations Manual Rev 0.6.32.*

### 1. What this document is

This is the theory-of-operations deep dive for the instrument as it actually ships at 0.6.32. It is factual. It describes the split between the Android APK and the Termux loopback services, why each model sits where it sits, how an utterance or a still becomes a log line, and what the unit will and will not do. The Field Operations Manual is the operator checklist. This file is the design argument.

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### 2. Thesis

A field instrument is useful when three things are true at once: the operator can point it, the unit can name what it pointed at, and the name can be spoken, stored, and shown to someone who does not share a language. Cloud multimodal suites do all three when a tower is up. They do none of them in a tent, a jungle track, or a basement. Mk I is the attempt to put that loop on one Snapdragon 888-class phone without a second computer.

The surprising piece is not any single model. Gemma-class chat, SmolVLM captions, SD Turbo stills, Kokoro speech, YAMNet, BirdNET, iNaturalist TFLite, and ML Kit translation all existed as separate apps or research checkpoints. The work was to keep four local HTTP servers alive together, route by spoken tuple instead of by app icon, and treat the phone sensors as part of the same instrument as the weights.

Image generation in thirty seconds to a few minutes is not the product. It is a pictogram factory for places with no library of pictures. Audio sense is not a toy classifier. It is a remote data logger that can sit in a tent while a second unit is at a fishing hole. Those two sentences are the design filter for every later feature.

### 3. System architecture

#### 3.1 Two process domains

The face is a Kotlin WebView APK (applicationId com.projecttricorder.app). The heavy generative weights live in Termux, not in the APK, because GGUF and ONNX graphs of this size do not belong inside an Android package and because Termux already has a working aarch64 toolchain. The APK talks to loopback only. If Termux is down, Inquiry, Sensor, Sentry, Imagine, and Kokoro go red. On-device TFLite, ML Kit, SpeechRecognizer, robot TTS, GNSS, Geocoder, and the log store keep working.

| Domain          | Owns                                                                                                                                                                                                                                 |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| APK / WebView   | Face, SpeechRecognizer, robot TTS, YAMNet, BirdNET, flora TFLite, ML Kit translate, field/audio/video logs, dictation JSON, PlaceStore, GnssClock, horizon IMU, Sol/sky canvases, Nearby mesh, OpenSky/NWS clients, knowledge cache. |
| Termux loopback | llama-server Gemma chat :8080, llama-server SmolVLM :8081, sd-server Turbo :8188, Kokoro ONNX :5000, sshd :8022.                                                                                                                     |

### 3.2 Why four ports stay up

Early builds tried to stop the image server to free RAM for chat. That was wrong. The operator wants chat, a still of a leaf, and a spoken caption in one session. The Termux launcher `~/bin/tricorder start` brings all four up and leaves them up. Health lamps CHAT / VISION / IMAGE / VOICE probe `/health` or the root of each port on a timer. Green idle, yellow in-flight, red unreachable.

### 3.3 Data path

Mic or typed line → JavaScript intercepts (dictation, clock, TRICORDER menu, SETTINGS, DIAGNOSTICS, COMPUTER, MEMORY ALPHA, MEDICAL, SUMMARIZE READINGS) → if nothing consumed it, the tuple engine assigns a mode and a model → fetch to 127.0.0.1 → CRT + optional Kokoro or robot TTS → FieldLog / AudioLog with a PlaceStore stamp. Vision loops skip the intercepts and go Sensor/Sentry → camera JPEG → SmolVLM → caption rules → log.

## 4. Tuple engine

The original design is a small keyword-and-mode router, not an agent framework. Button I seeds CHATTING. Button II seeds SCANNING. Button III seeds GUARDING. An Inquiry line that matches `imagine / show me / picture of` is tagged `IMAGE_GEN` and sent to `sd-server`. Everything else in CHATTING goes to Gemma. SCANNING always goes to SmolVLM on a dwell timer. GUARDING was first specified as motion-delta then escalate; that gated vision and was removed. Sentry now captions every dwell. The intercept layer in front of the router is where CLOCK, DICTATION, TRICORDER SENSOR, and the Enterprise easter egg live. Those never reach Gemma.

This split is load-bearing. If every wake word went to the LLM, dictation would become a chat reply and hands-free SENSOR would become a paragraph about sensors. The router is dumb on purpose.

## 5. Face map — every control

| Control           | What it does                                                                                                                                                  |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I Inquiry         | One-shot chat or Imagine. Mic is Android SpeechRecognizer. Text box optional (keyboard pop is a setting).                                                     |
| II Sensor         | Camera still on a dwell slider. SmolVLM caption. Flora pack may fire on the same JPEG. Hold is a toggle: freeze this frame until Hold is pressed again.       |
| III Sentry        | Same camera loop, continuous default. Caption watch for weapon / violence keywords. Robot-voice alerts, not Kokoro.                                           |
| IV Audio          | YAMNet scene classes on a listen/dwell pair of sliders. BirdNET when the pack is present. Context watch (NOAA, ADS-B, civic) if Wi-Fi and the checkbox is on. |
| T / XLATE         | ML Kit conversation. SOURCE and TARGET wheels. AUTO green swaps direction after each line.                                                                    |
| H Horizon         | IMU toy: heading, live G, GPS speed, baro altitude proxy, place line. Uncertified.                                                                            |
| S Sat/Stars       | TLE groups vs overhead star map. Toggle each press.                                                                                                           |
| O Sol             | Space weather + orbits + HF/VHF/UHF → earth forecast → moon phase → off.                                                                                      |
| L E V P           | Log review, officer audio log, video log, playback.                                                                                                           |
| Hold / Continuous | Hold freezes the current still. Continuous vision writes captions to the CRT without flooding Kokoro.                                                         |
| IPL / ALL         | Abort in-flight chat, vision, Imagine, voice, translate.                                                                                                      |
| STT slider        | Kill only the in-flight speech job.                                                                                                                           |
| ROBO              | Dark Kokoro → green robot voice → yellow [H/F R] hands-free menu.                                                                                             |

## 6. Speech in and speech out

### 6.1 Speech in

The APK uses Android SpeechRecognizer, not a Termux Whisper server. The config still lists whisper-tiny-en on :5001; that path was never the production mic. Recognizer results land in window.\_\_tricorderSpeechResult. Order of consumption: TRICORDER menu, clock (if open), dictation (if open), diagnostics/settings/backend, knowledge banks, readings, Inquiry.

### 6.2 Kokoro

Kokoro-82M ONNX, same family of weights PocketPal ships, served from Termux Python on :5000. The APK POSTs text and plays the returned WAV with MediaPlayer. Periods in USS and in sensor decimals were stripping utterances; a sanitizer removes those before the POST. Lines longer than 80 characters are chunked so the operator hears the start of the answer before the rest of the graph finishes. Kokoro is the measured voice. It is also the slow voice. Collisions with a second request produce silence or a late clip; IPL and the STT slider exist because of that.

### 6.3 Robot voice

Android TextToSpeech. Used for watch alerts (weapon, gunfire, humans, vehicles) because those must start now, and optionally for everything when ROBO is green or yellow. Flight announcements use robot voice so Kokoro is not occupied when the next ADS-B hit arrives.

### 6.4 Hands-free menu

Yellow [H/F R] keeps the recognizer armed. The grammar is TRICORDER plus a function name. Anything else is discarded so a kitchen conversation does not become an Inquiry. Audio, translator, clock, and dictation take the microphone while they run; the menu waits. This is not an always-on wake-word engine. It is a push-to-arm directory.

## 7. Inquiry and Imagine

Inquiry is Gemma on :8080, OpenAI-compatible /v1/chat/completions, modest context (launcher uses -c 2048, -t 3). The COMPUTER easter egg is intercepted before Gemma and spoken as a ship-status briefing (uptime, battery, baro, earth-link ping to 8.8.8.8). DIAGNOSTICS is the same idea with model and sensor health.

Imagine is SD Turbo Q4 on :8188 with --cfg-scale 1. Steps defaulted to the library 20 and wasted minutes on a 1–4 step checkpoint. The live trade is a pair of sliders: steps (4 is the confirmed-good floor; 2 is color squares; 6 is cleaner at about two minutes) and size 256 / 384 / 512. 256 at 4 steps was both slow and broken-looking; 512 at 4 steps recovered the cat. GPU offload on the Fold 3 Adreno path was judged not worth the build pain for this checkpoint. The field use is a pictogram you can hold next to a translated question.

## 8. Vision — Sensor and Sentry

Both modes take a camera JPEG on a dwell slider and POST it to SmolVLM2-500M-video-instruct on :8081. Sensor is the operator pointing at a thing. Sentry is the unit sitting on a rock. Hold freezes the JPEG so the caption is not rewritten every three seconds; that was flooding Kokoro. Continuous vision writes to the CRT only.

Flora/fauna is not a second camera. The same JPEG is handed to an on-device iNaturalist TFLite pack when the pack has been fetched. If the vision caption never mentioned a plant, the early builds never called the classifier. Current builds send the still regardless. A hit becomes a spoken flora line and a log kind flora.

Sentry watch is a word list on the caption: gun, knife, pistol, blade, and dual-use bats/pipes/chains. Sure weapon: klaxon and WEAPON DETECTED three times on robot voice. Possible / dual-use: seatbelt chime and a quieter line. Fighting, kicking, striking fire a possible-violence line. Motion-detected is spoken on robot voice and does not gate the caption loop. All hits land in the field log.

## 9. Audio sense

YAMNet TFLite is bundled in the APK. Audio Sense starts a listen/dwell duty cycle (two sliders). Each window is classified against AudioSet-style labels. A second pack, BirdNET, is downloaded at init when Earth link is up and identifies species when the score clears the pack threshold. The waveform on the CRT is the window just classified, not a continuous recorder.

Priority spoken watch, robot voice only: gunfire / artillery (klaxon, three times); scream / shout (conflict line); siren (emergency vehicles, three times); air / rail / ship; highway vehicles; named animal (dog detected); humans talking (three times); radio / TV / music. The unit mutes its own classification while it is speaking so a flight announcement does not become HUMANS DETECTED. Fireworks will look like gunfire. That is written on the warning page.

While Audio Sense is up and Wi-Fi reaches 8.8.8.8, ContextWatch polls the checked providers. NOAA CAP, OpenSky aircraft in a box around last GPS, and civic feeds if enabled. Defaults at 0.6.32: NOAA, aircraft, and citizen are ON. A remote Mk I on the Nearby mesh can receive the same line as remote tricorder unit detected (flight/weather/...) at HH:MM.

## 10. Overhead flight card

An OpenSky state in the GPS box that is airborne (not grounded) becomes a card: callsign, registration, type, owner, altitude and speed rewritten into feet/mph or meters/km/h per Settings, FROM / TO when adsbdb or OpenSky published a route. A JPEG is requested Planespotters → airport-data → JetPhotos and stored on the flight log snap. Touch the photo for an airframe facts card. Touch a city for Wikipedia + REST Countries + CIA Factbook. BACK returns the same craft. DISMISS AIRFRAME is required; the audio loop is not allowed to steal the picture.

## 11. Translator

ML Kit on-device packs, fetched at init if missing. SOURCE and TARGET are wheels, not sliders. AUTO green swaps after each completed line so a two-person conversation does not require a button. XLATE lamp: green waiting, red thinking, yellow speaking the translation. Silence ends the utterance. Reloading packs repairs a wedged recognizer session — that was a real 0.6.x bug and the reason the translator tears down and rebuilds the recognizer when the mode is re-entered.

## 12. Knowledge banks

MEMORY ALPHA is Simple English Wikipedia search plus extract and lead image, cached under app files. MEDICAL uses mdwiki.org only as a title catalog; body text is English Wikipedia because mdwiki REST is Cloudflare-blocked from the APK. Neither is Kiwix. A 30 MB illustrated Simple English ZIM does not exist; current illustrated ZIMs are hundreds of megabytes and are not auto-downloaded. Say memory alpha housefly or medical pink eye. TERMINATE leaves the bank. These are orientation layers, not a physician and not the Star Trek wiki.

## 13. Dictation

Wake word DICTATION (or TRICORDER DICTATION) opens a numbered directory of JSON memos, not a hidden chat. NEW asks for a title. EDIT nnnn enters the buffer. Spoken verbs: BACKSPACE, NEWLINE, PUNCTUATION PERIOD / COMMA / QUESTION, CURSOR MINUS word, CURSOR PLUS word, CURSOR CUT / CUT WORD, CURSOR PASTE, HOME, END. ZULU leaves the buffer menu:

SAVE or ABORT (ABORT requires CONFIRMED). SUMMARIZE nnnn writes a sibling memo titled summarized: %title%. MINDMAP keyword is a light semantic grouping (exact token plus close cousins such as puzzle / riddle). SPEAK nnnn reads the memo. KILL nnnn requires CONFIRMED. LOG nnnn appends the memo to the officer field log after CONFIRMED. TERMINATE returns to Inquiry. Every ten seconds the LED matrix restates the legal verbs for the current phase.

## 14. GNSS clock and places

CLOCK opens a face driven by Android GnssClock when FullBiasNanos is valid:  $\text{GPS time} = \text{TimeNanos} - \text{FullBiasNanos} - \text{BiasNanos}$ , then leap seconds to UTC. Display is interpolated to 4 Hz from a ~1 Hz hardware epoch. Fold 3 open-sky uncertainty is tens to a few hundred nanoseconds. That is a field clock. It is not a time standard and is not written into historical log timestamps. Until lock, device wall time is shown. Voice: analog, digital, twelve hour, twenty four hour, timer <label> <duration>, cancel, explain, terminate.

PlaceStore attaches to new log lines and to the horizon: GPS, Android Geocoder city/region when the geocoder has a name, and an offline Open Location Code (plus code). what3words is optional and needs a stored API key. Replay uses the stored string so a later review still names the site.

## 15. Sol, sky, horizon

Sol fetches space-weather and propagation summaries at init when Earth link is up, plus a local forecast. Press 1: orbits and space weather and HF/VHF/UHF (UHF aurora on its own line). Press 2: about six hours of local weather and a three-day outlook, scrolling. Press 3: moon phase from a 29.53058867-day synodic month, drawn on canvas. Press 4: off.

Sat/Stars uses a TLE snapshot for amateur and major comms constellations and a star map tied to device attitude and time. Horizon is SensorManager pitch/roll/yaw plus computed G from the accelerometer, GPS speed, and barometric altitude proxy. Attitude needs the phone in the orientation the IMU fusion expects; early builds required a 90-degree rotate. Uncertified. Must never fly on it.

## 16. Readings and logs

Every watch hit, sensor still, sentry still, flora line, flight card, officer memo, and video clip is a JSON object with time, kind, text, optional JPEG/WAV/MP4, GPS, and place string. Log review is a time slider. SUMMARIZE READINGS [category] [day|week|month|all] draws letter-labeled color bars with QTY, a legend, a text tally, and a filtered timeline when a bar is touched. Categories include vehicles, entertainment, flights, species, birdsong, humans, speech, weapons, and the other watch kinds.

Video log is the front camera. Officer log is a toggled spoken memo. Playback of audio and video entries was added after the timeline could list them but could not open them.

## 17. Mesh

Nearby Connections / Wi-Fi Aware style P2P carries watch alerts between two Mk I units. The documented pair is a tent unit on Audio Sense and a fishing-hole unit. The remote line is spoken and logged. This is an opt-in local link, not a cloud panopticon. There is no account server.

## 18. Initialization

On first launch with Earth link (8.8.8.8 reachable) the APK says initializing tricorder and fetches BirdNET, flora/fauna TFLite, and ML Kit language packs if they are absent. It does not fetch Gemma, SmolVLM, SD Turbo, or Kokoro weights — those stay a Termux procedure documented in TERMUX-BACKEND-INSTALL.md. If Termux is missing, the face warns. Health lamps stay red until ~/bin/tricorder start reports the four ports up.

## 19. Why the pieces are split the way they are

| Job            | Why this engine                                                                  |
|----------------|----------------------------------------------------------------------------------|
| Chat           | Gemma-class GGUF on llama-server. Small enough for 888 RAM if context is modest. |
| Scene caption  | SmolVLM 500M. Fast enough for a dwell loop. Not a detector.                      |
| Pictogram      | SD Turbo Q4, cfg-scale 1, 4–6 steps. Seconds-to-minutes, not interactive art.    |
| Measured voice | Kokoro ONNX. Quality over speed. Chunked.                                        |
| Alert voice    | Android TTS. Immediate.                                                          |
| Mic            | Platform SpeechRecognizer. No extra daemon.                                      |
| Scene audio    | YAMNet in-APK. No Termux.                                                        |
| Bird species   | BirdNET pack fetched once.                                                       |
| Flora          | iNaturalist TFLite on the Sensor JPEG.                                           |
| Translate      | ML Kit packs. Offline conversation.                                              |
| Time           | GnssClock when locked, else device clock.                                        |
| Place          | Geocoder + plus codes offline.                                                   |

Cloud vendors have each of these as a service. They do not ship them as one offline appliance because the product incentive is an account, a quota, and a tower. Open-source phone LLM apps optimize for a single GGUF and a clean chat UI. ADS-B and planetarium apps optimize for one sensor domain. Mk I is the awkward union. That is why it looks like a tricorder instead of a chatbot.

## 20. Capabilities at 0.6.32

| Capability                  | Offline?          | Notes                                  |
|-----------------------------|-------------------|----------------------------------------|
| Spoken chat (Gemma)         | Yes, if Termux up | Inquiry.                               |
| Scene caption (SmolVLM)     | Yes, if Termux up | Sensor / Sentry.                       |
| Still generation (SD Turbo) | Yes, if Termux up | Steps/size sliders.                    |
| Measured TTS (Kokoro)       | Yes, if Termux up | Chunked >80 chars.                     |
| Robot TTS                   | Yes               | Watch + ROBO.                          |
| Hands-free directory        | Yes               | Yellow [H/F R].                        |
| YAMNet watch                | Yes               | In APK.                                |
| BirdNET species             | Yes after fetch   | Needs one Earth-link init.             |
| Flora/fauna TFLite          | Yes after fetch   | Same.                                  |
| ML Kit translate            | Yes after fetch   | 50+ packs.                             |
| MEMORY ALPHA / MEDICAL      | Fetch per article | Cached afterward.                      |
| ADS-B card + photo          | Needs link        | OpenSky / photo CDNs.                  |
| NOAA / civic alerts         | Needs link        | Audio Sense + checkbox.                |
| Mesh alerts                 | Local radio       | Two units.                             |
| GNSS clock                  | Yes with sky      | Novelty timing.                        |
| Plus-code places            | Yes               | Geocoder names need a prior fix.       |
| Moon / Sol / stars          | Mixed             | Moon on-device; space weather at init. |
| Dictation + mindmap         | Yes               | JSON memos.                            |
| Readings charts             | Yes               | From the log store.                    |

## 21. Hard limits

Not FAA/NASA certified. Horizon, G, speed, baro altitude, satellites, stars, Sol, moon, and GNSS clock must never fly a craft. Weapon and gunfire lines are captions and AudioSet labels. MEDICAL is an encyclopedia extract. Image generation is slow and can tile or collapse at 256 or at 2 steps. Kokoro will drop or delay under collision. Hands-free is not a far-field wake word. Two units on mesh is a pair, not a network of record. Snapdragon 888 RAM is the ceiling; four Termux servers plus the APK will die if a fifth giant graph is added without dropping one of the four.

## 22. Theory of the operator loop

Point. Name. Speak. Store. Show. Those five verbs are the instrument. Sensor is point and name. Kokoro or ROBO is speak. FieldLog is store. Imagine plus translator is show, including to someone who does not read the caption. Sentry and Audio Sense run the same five verbs on a timer so the operator can put the unit down. The hands-free menu is only how the operator picks which verb without looking at the face. If a proposed feature does not serve one of those five, it does not belong on Mk I.

## 23. Related documents

| File                                                | Role                                                  |
|-----------------------------------------------------|-------------------------------------------------------|
| Project-Tricorder-MkI-Operations-Manual-0.6.32.docx | Operator manual, warnings, troubleshooting.           |
| android-app/.../www/manual.html                     | On-device short briefing.                             |
| docs/TERMUX-BACKEND-INSTALL.md                      | How to get the four servers up.                       |
| termux/tricorder                                    | Launcher: start / status / stop + sshd 8022.          |
| This file                                           | Features, capabilities, theory of operations, 0.6.32. |

End of the 0.6.32 complete guide. Designed and coded in the USA by Jim Ames, Claude and Grok. N2NHU Labs, Newburgh, New York 12550.